CALVIN GACHERU MWANGI

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Portfolio: https://calvin-gacheru.github.io/MyPortfolio/

PROFESSIONAL SUMMARY

Data Scientist, AI Engineer, and full-stack developer specializing in the complete software lifecycle. Experienced in building Retrieval-Augmented Generation (RAG) systems, training predictive machine learning models, and managing cloud deployments. Focuses on shipping scalable, production-ready solutions and enterprise ETL pipelines end-to-end.

TECHNICAL SKILLS

- **Languages:** Python, SQL, HTML/CSS/JavaScript, Bash, R
- **AI & Machine Learning:** scikit-learn, LangChain, LangGraph, RAG, Agentic Orchestration, Vector DBs (Qdrant), OpenAI API, Groq LLMs, Pandas, NumPy, NLP, Time Series Analysis
- **Infrastructure & Deployment:** Docker, Kubernetes, Git/GitHub Actions (CI/CD), Linux/Fedora, Render
- **Web Frameworks & Databases:** FastAPI, Flask, Django, PostgreSQL, SQLite, Redis

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EXPERIENCE

BikBat Insurance Agency | Freelance Web Developer | 2026

- Engineered and launched a complete corporate brochure website for a commercial client.
- Managed end-to-end deployment, including domain registration, DNS configuration, and cloud hosting.

ProdigyInfoTech | Data Science Virtual Intern | 2025

- Completed 4 end-to-end data science projects utilizing real-world datasets.
- Executed exploratory data analysis, predictive modeling, customer churn analytics, and sentiment analysis with fully documented, reproducible workflows.

KEY PROJECTS

MySaidi - AI Productivity Assistant

- Developed a full-stack Progressive Web App (PWA) using FastAPI, PostgreSQL, and a vanilla JavaScript frontend.
- Integrated LangGraph and Groq LLMs to create an agentic AI chat assistant capable of managing schedules, tasks, and habits via natural language.
- Implemented secure JWT authentication and a responsive, mobile-optimized user interface.

Uber Price Prediction & Production MLOps Pipeline

- Built an enterprise-grade ML pipeline predicting Uber ride prices using an optimized Random Forest Regressor.
- Designed a multi-container orchestration system using Docker and automated CI/CD workflows via GitHub Actions.
- Served the inference model through a FastAPI REST backend with Pydantic validation and an interactive Streamlit UI.

Behavioural Explainer RAG System

- Engineered a production-grade Retrieval-Augmented Generation (RAG) tool analyzing human interactions using psychological frameworks.
- Built an ingestion pipeline for PDF documents utilizing PyMuPDF and stored vector embeddings in Qdrant.
- Utilized Llama 3.1 and Ollama for local LLM inference, operated via a custom Command Line Interface (Typer).

Maternal Risk Detection System

- Trained predictive models to classify maternal health risks from patient vital indicators.
- Integrated SHAP (Explainable AI) to flag real-time outliers and build clinical trust in the model's predictions.
- Automated patient SMS reminders via Africa's Talking API and APScheduler, managing records in a local SQLite database.

ETL Data Engineering & Custom Web Apps

- **ETL Pipelines:** Built custom processes for automated data extraction, transformation, and loading using Python and SQL.
- **Arcade Kenya & Valen Time Love:** Deployed interactive platforms and custom web applications utilizing HTML5 Canvas, CSS3, and JavaScript.

EDUCATION

United States International University — Africa (USIU-Africa) | Nairobi, Kenya

BSc. Data Science & Analytics | Expected Graduation: 2027

CERTIFICATIONS

- **Building AI Agents using LangGraph** – Simplilearn (Focus on stateful orchestration and LLM applications)
- **Data Science Virtual Internship** – ProdigyInfoTech